

User Monitor - Product Analysis Report

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1. Product Overview

Architecture

Component	Technology	Purpose
Web Dashboard	React + Vite + Tailwind CSS + Shadcn UI	Admin/Manager/User web interface
Backend API	Node.js + Express + PostgreSQL	REST API, authentication, data storage
Desktop Agent	Electron + better-sqlite3	Activity monitoring, screenshots, app tracking
Browser Extension	Chrome MV3 + Firefox MV2	Browser domain/tab tracking
Real-time	WebSocket (ws)	Live heartbeat & status updates

Current Feature Set

- **Authentication & Authorization:** JWT-based auth, role-based access (orgadmin, manager, user), forgot/reset password
- **User Management:** CRUD, manager hierarchy, force logout, password reset by admin/manager, per-user feature overrides
- **Organization Settings:** Timezone, shift times, work days, feature toggles (screenshots, AFK, breaks, activity tracking)
- **Activity Monitoring:** Keyboard/mouse event tracking, idle/AFK detection with configurable thresholds
- **Screenshots:** Periodic capture with configurable intervals, upload and storage, viewable in reports
- **Work Sessions:** Start/end tracking, total work/idle/break seconds, status management
- **App Tracking:** Desktop application usage logging, app categorization (productive/neutral/non-productive), productivity scoring

- **Browser Tracking:** Browser extension for domain-level tracking, native messaging + HTTP fallback, expandable browser domain view in App Usage dashboard
- **Break Management:** Configurable break types with limits, break logging, violation notifications to managers
- **Dashboard & Stats:** Admin/manager/user dashboards, hourly activity stats, team comparison, productivity scores
- **Reports:** Daily summary, break usage, screenshots, idle events, PDF export
- **Timeline:** Visual activity timeline per user per day
- **Audit Logs:** Comprehensive audit trail with actor, action, target, old vs new values, IP address
- **Notifications:** In-app notifications (break violations, etc.), notification bell with mark-as-read
- **Multi-tenant:** Full org_id isolation across all tables

Desktop Agent Capabilities

- Auto-login with stored credentials
- Activity monitoring (keyboard/mouse events)
- Screenshot capture at configurable intervals
- App usage tracking (active window detection)
- Browser detection (Chrome, Edge, Brave, Opera, Firefox)
- Silent browser extension installation (Windows Registry / Linux policies)
- Periodic browser re-scanning for newly installed browsers (every 6 hours)
- Local SQLite storage with batch sync to server every 5 minutes
- Heartbeat system for online/offline status
- Force logout support from server
- Power save blocker to prevent system sleep during tracking

Database Tables

Table	Partitioned	Description
organizations	No	Company details and subscription limits
org_features	No	Organization-level feature toggles
users	No	User accounts with role and manager hierarchy
user_features	No	Per-user feature overrides
break_master	No	Break type definitions with limits
work_sessions	No	Work session start/end with totals
agent_sessions	No	Device/agent session tracking
activity_logs	Yes (monthly)	High-volume keyboard/mouse activity logs
screenshots	No	Screenshot metadata and storage paths
break_logs	No	Break start/end/duration records
audit_logs	No	Change audit trail with old/new values
heartbeats	No	Agent heartbeat history

Table	Partitioned	Description
notifications	No	In-app notification records
app_categories	No	Application categorization definitions
tracked_apps	No	Discovered application registry
app_usage_logs	Yes (monthly)	High-volume app usage time logs
user_app_summary	No	Aggregated daily app usage stats
browser_activity_logs	No	Browser domain/tab activity from extensions

2. Competitive Analysis

Feature Comparison with Market Leaders

Feature	Time Doctor	Hubstaff	ActivTrak	Teramind	User Monitor
Activity tracking	Yes	Yes	Yes	Yes	Yes
Screenshots	Yes	Yes	No	Yes	Yes
App tracking	Yes	Yes	Yes	Yes	Yes
Browser/URL tracking	Yes	Yes	Yes	Yes	Yes
Break management	Basic	Basic	No	No	Yes (Advanced)
Idle/AFK detection	Yes	Yes	Yes	Yes	Yes
Productivity scoring	Yes	Yes	Yes	Yes	Yes
PDF reports	Some	Some	Yes	Yes	Yes
Audit logs	No	No	Yes	Yes	Yes
Self-hosted option	No	No	No	Yes	Yes
Per-user feature toggles	No	No	No	Limited	Yes
Multi-org/tenant	Yes	Yes	Yes	Yes	Yes
Project/Task tracking	Yes	Yes	No	No	No
GPS/Location tracking	No	Yes	No	No	No
Payroll integration	Yes	Yes	No	No	No
Client/Invoice reporting	Yes	No	No	No	No
Live screen viewing	Yes	No	No	Yes	No
Mobile app	Yes	Yes	No	No	No
Alert rules engine	Limited	Limited	Yes	Yes	No
SSO/SAML	Yes	Yes	Yes	Yes	No

Current Differentiators

- 1. **Self-hosted option** - Companies with strict data policies can run it on their own infrastructure. Most competitors are cloud-only.
- 2. **Advanced break management** - Configurable break types with time limits, usage tracking, and automatic violation notifications to managers. Most competitors offer only basic pause/resume.
- 3. **Per-user feature overrides** - Disable screenshots for HR but enable for developers. Granular control at the individual level.
- 4. **Comprehensive audit trail** - Full change history with who, what, when, old values, new values. Important for compliance-heavy industries.
- 5. **Hybrid browser tracking** - Extension-based for accuracy + window title fallback for any browser. Future-proof against new browsers.

3. SaaS Market Feature Comparison

What's Built (Core SaaS Features)

Category	Status	Details
Authentication	Done	JWT, role-based, forgot/reset password
Multi-tenancy	Done	org_id isolation on all tables
User management	Done	CRUD, hierarchy, force logout
Activity monitoring	Done	Keyboard/mouse events, idle detection
Screenshots	Done	Periodic capture, configurable intervals
App tracking	Done	Desktop app usage + productivity categorization
Browser tracking	Done	Extension + hybrid approach
Break management	Done	Types, limits, violations, notifications
Dashboard	Done	Admin/manager/user views
Reports	Done	Summary, breaks, screenshots, idle, PDF export
Audit logs	Done	Full change trail with old/new values
Notifications	Done	In-app with bell icon
Real-time	Partial	WebSocket for heartbeats, but dashboard doesn't auto-refresh

What's Missing for Production SaaS

Category	Feature	Priority	Details
Billing	Subscription/payment system	Critical	No Stripe/Razorpay integration, no pricing tiers, no trial management

Category	Feature	Priority	Details
Onboarding	Welcome wizard & guided setup	High	New orgs have no guided first-time experience
Teams	Department/team structure	High	Only manager hierarchy, no department grouping
Email	Transactional emails	High	No welcome emails, alert emails, or weekly digest
Agent Updates	Auto-update mechanism	High	No way to push agent updates to installed clients
Super Admin	Platform admin panel	Medium	No way to manage all organizations from a single admin view
Rate Limiting	Per-org API limits	Medium	Basic rate limiting exists but not org-aware
Data Export	CSV/Excel export	Medium	Only PDF export currently available
Webhooks	Event notification system	Medium	No way for orgs to integrate with external tools
SSO/SAML	Single Sign-On	Medium	Enterprise customers expect this
2FA	Two-factor authentication	Medium	No additional auth security layer
Mobile App	iOS/Android	Low	Not critical for initial launch
White-labeling	Custom branding	Low	Not needed initially, useful for resellers
API Docs	Swagger/OpenAPI documentation	Low	No public API documentation

4. Buyer Perspective

Questions a Buyer Would Ask

Security & Compliance

Question	Current Answer	Gap
"Where is my data stored?"	Can be self-hosted or cloud	Good - this is a selling point
"Is data encrypted at rest?"	No encryption at rest	Need to add PostgreSQL encryption or disk-level encryption
"Is data encrypted in transit?"	HTTPS for API, but agent uses HTTP in dev	Need HTTPS enforcement in production
"Do you have SOC 2 / ISO 27001?"	No	Long-term goal for enterprise sales

Question	Current Answer	Gap
"Can employees see what's being tracked?"	Yes, employees have their own dashboard	Good - transparency builds trust
"GDPR compliance?"	No specific GDPR features	Need data deletion requests, consent management

Functionality

Question	Current Answer	Gap
"Can I set rules per department?"	Per-user overrides, but no departments	Need department/team structure
"Can I integrate with Slack/Teams?"	No integrations	Need webhook/integration system
"Can I export to PowerBI/Tableau?"	PDF only	Need CSV/Excel/API export
"Does it work on Mac?"	No, Windows + Linux only	Mac support needed for many companies
"What if agent loses internet?"	Stores locally, syncs when back online	Good - already handled
"Can I track project time?"	No project/task system	Need project time tracking

Privacy

Question	Current Answer	Gap
"Can employees opt out of screenshots?"	Admin can disable per-user	Good
"Is there a personal time / privacy mode?"	No	Need a way for users to pause tracking during personal time
"Can we blur sensitive content in screenshots?"	No	Useful for banking/personal email protection
"Do you track keystrokes content?"	No, only event counts	Good - privacy-friendly approach

Pricing & Business

Question	Current Answer	Gap
"Per user per month pricing?"	No billing system	Critical - need pricing tiers
"Free trial?"	No trial system	Need trial period with auto-expiry

Question	Current Answer	Gap
"What's in each tier?"	No tiers defined	Need Basic/Pro/Enterprise tiers
"Annual discount?"	No billing	Need annual billing option

Key Suggestions from a Buyer's Perspective

1. **Employee transparency dashboard** - Give employees a clean "My Day" view showing their own productivity, breaks, and stats. Builds trust and reduces resistance to monitoring.
2. **Screenshot blurring** - Option to automatically blur sensitive areas or let admins choose which apps trigger blurring.
3. **Working hours enforcement** - Agent should respect shift hours and not track outside them (or mark it differently).
4. **Data retention settings** - Let admins choose: keep data for 30/60/90/365 days. Important for storage costs and compliance.
5. **Quick onboarding** - "Get started in 5 minutes" experience: create org, invite users, download agent, start tracking.
6. **Comparison with self** - Show employees/managers "this week vs last week" trends, not just raw numbers.

5. Timezone & Multi-Shift Architecture

Current Implementation

Component	How it works
Organization timezone	<code>organizations.timezone</code> (e.g., <code>Asia/Kolkata</code>)
User timezone	<code>users.timezone</code> (per user, overrides org)
Shift configuration	Single shift per org: <code>shift_start_time</code> and <code>shift_end_time</code>
Work date calculation	Uses user's timezone: <code>AT TIME ZONE user.timezone</code>
Reports	Date-based filtering using <code>work_date</code> column

What Works

- Single timezone org with one shift
- Users in different timezones within same org (each user has their own timezone)
- Work date calculation respects individual user timezone

What Doesn't Work

Problem	Details
Only 1 shift per org	Can't define Day shift (9am-6pm) AND Night shift (10pm-7am)
No shift assignment per user	Can't say "User A is Day shift, User B is Night shift"

Problem	Details
Night shift crossing midnight	Shift from 10pm to 7am spans two calendar dates, <code>work_date</code> might split it
No shift rotation	Can't configure "Week 1: Day shift, Week 2: Night shift"
Reports are date-based	Night shift 10pm Feb 15 to 7am Feb 16 - which date does it count as?

Recommended Architecture for Multi-Shift

New Tables Needed

```
shift_master
- id UUID
- org_id UUID
- name VARCHAR (e.g., "Day Shift", "Night Shift", "US East Shift")
- start_time TIME
- end_time TIME
- timezone VARCHAR
- work_days TEXT[] (e.g., ["Mon","Tue","Wed","Thu","Fri"])
- is_active BOOLEAN

user_shifts
- id UUID
- user_id UUID
- shift_id UUID
- effective_from DATE
- effective_to DATE (NULL = current)
```

How Reports Would Work

- 1. Get user's assigned shift
- 2. Use shift's timezone and start/end times
- 3. For night shifts crossing midnight: the `work_date` is the date when the shift STARTS
- 4. Example: Night shift 10pm Feb 15 to 7am Feb 16 → `work_date` = Feb 15
- 5. Reports filter by shift-aware `work_date`, not just calendar date

Multi-timezone Organization Example

```
Organization: Global Corp (timezone: UTC)

Shift 1: "India Day"      - 9:00 AM to 6:00 PM IST (Asia/Kolkata)
Shift 2: "US East"       - 9:00 AM to 6:00 PM EST (America/New_York)
Shift 3: "India Night"   - 10:00 PM to 7:00 AM IST (Asia/Kolkata)

User A → India Day shift → Reports calculated in IST
User B → US East shift → Reports calculated in EST
User C → India Night shift → Reports use shift-start date as work_date
```


6. Data & UX Audit

Data Stored but NOT Visible in UI

Data	Stored In	Visible in UI?	Action Needed
Detailed activity logs (keyboard/mouse events)	activity_logs	Only as summary stats	Show detailed activity timeline
Heartbeat history	heartbeats	Only as online/offline dot	Could show "last seen" history
Agent sessions (device info)	agent_sessions	Not shown anywhere	Show devices per user
Browser activity (domains)	browser_activity_logs	Only via expandable table row	Needs dedicated charts/visualization
Notification history	notifications	Bell dropdown only	Need full notification history page
App category productivity breakdown	user_app_summary	In dashboard	Could have more detail
Audit log details	audit_logs	Activity Logs page	Good - already shown
Screenshot metadata	screenshots	In Reports	Could improve gallery view

UX Issues & Improvements

Issue	Current State	Improvement
No quick date filters	User must manually pick start/end dates	Add "Today / This Week / Last 7 Days / This Month" buttons
No auto-refresh	Dashboard requires manual reload	Use WebSocket (already have infrastructure) to auto-update
Timeline hidden	Timeline page exists but commented out of sidebar	Enable in sidebar navigation
No "My Day" view	Regular users see limited data	Create a clean employee self-view page
No trend comparison	Only shows current period data	Add "vs last week" or "vs last month" comparison

Issue	Current State	Improvement
No search in app usage	Must scroll through apps	Add search/filter in Top Applications table
No browser activity charts	Only table view for domains	Add pie chart for top domains, bar chart for time by domain
No empty state guidance	Pages show "No data" without help	Add "Get started" messages with setup instructions
No mobile responsiveness	Dashboard may break on mobile	Add responsive design for tablets at minimum
No keyboard shortcuts	All mouse-driven	Add shortcuts for common actions

User Flow Issues

Flow	Problem	Fix
New org setup	After register, user lands on empty dashboard with no guidance	Add onboarding wizard
Agent download	No in-app link to download the agent	Add "Download Agent" button in settings/dashboard
First-time user	New employee doesn't know what's being tracked	Add "What we track" info panel
Manager view	Manager sees all users but can't quickly compare	Team comparison page exists but could be more visual
Notification actions	Notifications show info but no action buttons	Add "View User" or "View Details" links in notifications

7. Priority Action Items

Critical (Do Before Launch)

#	Task	Effort	Impact
1	Fix partition auto-creation (activity_logs breaks July 2026, app_usage_logs breaks June 2026)	1 day	System will crash without this
2	Add billing/subscription system (Stripe/Razorpay)	1-2 weeks	Can't monetize without it
3	Add transactional email system (welcome, password reset, alerts)	3-5 days	Basic SaaS requirement
4	HTTPS enforcement for production	1 day	Security requirement

#	Task	Effort	Impact
5	Agent auto-update mechanism	3-5 days	Can't push fixes to deployed agents

High Priority (First Month After Launch)

#	Task	Effort	Impact
6	Implement hybrid browser tracking (window title fallback)	2-3 days	Future-proof browser support
7	Add department/team structure	3-5 days	Better organization management
8	Multi-shift support with user-shift assignment	1 week	Required for 24/7 operations
9	Quick date filters (Today, This Week, etc.)	1 day	Major UX improvement
10	Dashboard auto-refresh via WebSocket	2 days	Real-time feel
11	CSV/Excel data export	2-3 days	Common customer request
12	New org onboarding wizard	3 days	Better first-time experience

Medium Priority (First Quarter)

#	Task	Effort	Impact
13	Screenshot blurring option	3-5 days	Privacy compliance
14	Two-factor authentication (2FA)	3 days	Security enhancement
15	Privacy mode / personal time toggle	2 days	Employee trust
16	Data retention settings (auto-delete old data)	2 days	Storage management
17	Webhook system for integrations	1 week	Enables Slack/Teams integration
18	Weekly email digest for managers	3 days	Keeps managers engaged
19	Super admin panel for platform management	1 week	Multi-org management
20	API rate limiting per organization	2 days	Fair usage enforcement

Low Priority (Future Roadmap)

#	Task	Effort	Impact
21	Mobile app (iOS/Android)	2-3 months	Field worker tracking
22	SSO/SAML integration	2 weeks	Enterprise sales requirement
23	Mac agent support	2-3 weeks	Broader market

#	Task	Effort	Impact
24	Project/task time tracking	2-3 weeks	Competes with Time Doctor
25	Live screen viewing	2 weeks	Enterprise feature
26	White-labeling	1 week	Reseller channel
27	Public API documentation (Swagger)	3 days	Developer ecosystem
28	Alert rules engine	2 weeks	Custom monitoring rules

Appendix: Technology Stack

Frontend

- React 18 + Vite
- Tailwind CSS + Shadcn UI (Radix primitives)
- Zustand (state management)
- React Hook Form + Zod (forms/validation)
- Recharts (data visualization)
- Axios (HTTP client)
- Sonner (toast notifications)
- Lucide React (icons)
- date-fns + date-fns-tz (date handling)

Backend

- Node.js + Express
- PostgreSQL (with table partitioning)
- JWT (authentication)
- Bcrypt (password hashing)
- WebSocket (ws) for real-time
- Multer (file uploads)
- PDFKit (PDF generation)
- Zod (validation)
- Helmet + CORS (security)
- Morgan (logging)
- node-schedule (scheduled tasks)
- express-rate-limit (rate limiting)

Desktop Agent

- Electron 28
- better-sqlite3 (local storage)
- active-win (active window detection)
- uiohook-napi (keyboard/mouse events)
- screenshot-desktop (screen capture)
- electron-store (config persistence)

- Winston (logging)
- Luxon (date/time handling)
- electron-builder (packaging: NSIS for Windows, AppImage for Linux)

Browser Extension

- Chromium: Manifest V3 (Chrome, Edge, Brave, Opera)
- Firefox: Manifest V2
- Native Messaging + HTTP fallback
- Background service worker

This document was generated as part of a product analysis for User Monitor v1.0.0.